■ EXCEL NEW FORMULAS TASK BOOK

4 Hands-On Tasks · ~2 Hours · Class 10 Level
Learn formulas you've NEVER used before — with real data practice

TASK	FORMULAS YOU LEARN	TIME
Task 1 — Student Report Card	AVERAGE, MAX, MIN, RANK	~25 min
Task 2 — Shop Sales Analysis	SUMIF, COUNTIF, AVERAGEIF	~30 min
Task 3 — Employee Details Cleanup	TRIM, PROPER, CONCATENATE, MID, RIGHT	~30 min
Task 4 — Date & Time Calculations	TODAY, NOW, DATEDIF, EOMONTH, WEEKDAY	~30 min

Before You Start: Open Microsoft Excel → New Blank Workbook → Create a new sheet for each task. Type all data exactly as shown. The formula column is where YOU write the formula. Every formula starts with the = sign. Take your time — understanding is more important than speed!

■ TASK 1 — Student Report Card (≈25 minutes)

■ Scenario

You are a class teacher. You have marks of 8 students in 4 subjects. You need to find the average marks, highest marks, lowest marks, and rank each student in the class. This is exactly how school report cards are made!

■ Step 1: Type This Data in Excel (Sheet name: Report Card)

A	B	C	D	E	F	G	H	I
Name	Maths	Science	English	Hindi	Total	Average	Rank	Grade
Ravi	78	85	72	90				
Priya	92	88	95	87				
Aman	65	70	68	75				
Sunita	88	92	80	85				
Rohit	55	60	58	62				
Kavya	95	98	92	96				
Deepak	72	68	75	70				
Meena	80	75	82	78				
Highest								
Lowest								

■ Yellow columns = where you write formulas | Red rows = class highest & lowest

■ Step 2: Learn These 4 Formulas

■ AVERAGE — Finds the mean (average) of numbers

```
=AVERAGE(B2, C2, D2, E2)
```

In cell G2 type: =AVERAGE(B2:E2)

This adds B2+C2+D2+E2 and divides by 4 automatically.

Copy this formula down to G3, G4... G9 for all students.

■ MAX — Finds the HIGHEST number in a range

```
=MAX(G2:G9)
```

In cell G11 (Highest row) type: =MAX(G2:G9)

This finds the highest average in the class.

Also use: =MAX(B2:B9) to find highest Maths marks.

■ MIN — Finds the LOWEST number in a range

```
=MIN(G2:G9)
```

In cell G12 (Lowest row) type: =MIN(G2:G9)

This finds the lowest average in the class.

Minimum marks help identify students who need extra help.

■ RANK — Gives position/rank among all numbers (1 = highest)

```
=RANK(G2, $G$2:$G$9, 0)
```

In cell H2 type: =RANK(G2, $G$2:$G$9, 0)

G2 = the number to rank | $G$2:$G$9 = full list ($ locks it)

0 = rank from highest to lowest (1 = topper)

Copy this formula down for all students.

■ Step 3: Bonus — Add a Grade column

In cell I2 type: =IF(G2>=90,"A+",IF(G2>=80,"A",IF(G2>=70,"B",IF(G2>=60,"C","D")))) This uses nested IF to give grades. Copy down for all students.

> ■ *WHY THE $ SIGN IN RANK? When you copy a formula down, Excel automatically changes row numbers. G2 becomes G3, G4, etc. — that is GOOD for the student's score. But the full range ($G$2:$G$9) must stay FIXED for all students. The $ sign tells Excel: 'don't change this!' — called an Absolute Reference.*

■ TASK 2 — Shop Sales Analysis (≈30 minutes)

■ Scenario

You run a mobile accessories shop. You have a list of 12 sales transactions. You want to find: total sales of a specific item, how many times each item was sold, and the average selling price per item. These are the SUMIF, COUNTIF, and AVERAGEIF formulas — the upgraded versions of SUM, COUNT, and AVERAGE!

■ Step 1: Type This Data (Sheet name: Shop Sales)

A	B	C	D
Sale Date	**Item Name**	**Quantity**	**Sale Amount (Rs)**
01-Jun-25	Mobile Cover	3	600
02-Jun-25	USB Cable	5	750
03-Jun-25	Mobile Cover	2	400
04-Jun-25	Screen Protector	4	480
05-Jun-25	USB Cable	3	450

06-Jun-25	Wall Charger	2	600
08-Jun-25	Mobile Cover	6	1200
10-Jun-25	Screen Protector	2	240
12-Jun-25	USB Cable	4	600
14-Jun-25	Wall Charger	3	900
16-Jun-25	Mobile Cover	1	200
18-Jun-25	Screen Protector	5	600

■ Step 2: Create a Summary Table below your data (Row 16 onwards)

F	G	H	I
Item Name	**Total Sales Rs (SUMIF)**	**Times Sold (COUNTIF)**	**Avg Price Rs (AVERAGEIF)**
Mobile Cover	You write formula	You write formula	You write formula
USB Cable	You write formula	You write formula	You write formula
Screen Protector	You write formula	You write formula	You write formula
Wall Charger	You write formula	You write formula	You write formula

■ Step 3: Learn These 3 Formulas

■ SUMIF — Adds amounts ONLY for a specific item/condition

```
=SUMIF(B2:B13, F2, D2:D13)
```

B2:B13 = column to check (Item Name column)

F2 = the item to look for (Mobile Cover)

D2:D13 = column to add (Sale Amount)

Result: total rupees earned from Mobile Cover sales only.

Copy formula for USB Cable, Screen Protector, Wall Charger.

■ COUNTIF — Counts rows ONLY where a condition is met

```
=COUNTIF(B2:B13, F2)
```

B2:B13 = column to check (Item Name column)

F2 = the item to count (Mobile Cover)

Result: how many times Mobile Cover was sold (answer = 4 times).

Useful for: how many transactions? how many customers?

■ AVERAGEIF — Finds average ONLY for rows matching a condition

```
=AVERAGEIF(B2:B13, F2, D2:D13)
```

B2:B13 = column to check | F2 = item to match | D2:D13 = amounts

Result: average sale amount per Mobile Cover transaction.

This tells you: on average, how much do customers spend on each item?

■ TALLY CONNECTION: In Tally, the Sales Register auto-calculates item-wise totals using this exact SUMIF logic internally. When you see 'Mobile Cover — Total: Rs 2,400' in Tally's Item-wise Sales report, that's a SUMIF on all F8 Sales vouchers!

■ TASK 3 — Employee Data Cleanup (≈30 minutes)

■ Scenario

Your HR team gave you messy employee data — names are in WRONG case, some have extra spaces, and you need to split full names and combine information. This is called Data Cleaning — one of the most common real-world Excel tasks done by every company's HR and accounts team!

■ Step 1: Type This Messy Data (Sheet name: Employee Data)

A	B	C	D
Employee ID	**Full Name (Messy)**	**Department Code**	**Email (lowercase)**
EMP001	ravi kumar	SALES-01	ravi.kumar@company.com
EMP002	PRIYA SINGH	ACCT-02	priya.singh@company.com
EMP003	aman verma	WARE-03	aman.verma@company.com
EMP004	SUNITA RAO	ADMN-04	sunita.rao@company.com
EMP005	rohit sharma	SALE-05	rohit.sharma@company.com
EMP006	KAVYA PATEL	MKTG-06	kavya.patel@company.com

■ Notice: extra spaces, ALL CAPS, all lowercase, mixed — all messy!

■ Step 2: Create Clean Output Columns (Column F onwards)

F = Clean Name | G = First Name Only | H = Last Name Only | I = Dept Only | J = Full Info Combined

■ Step 3: Learn These 5 Text Formulas

■ TRIM — Removes all extra spaces from text (before, after, and between words)

```
=TRIM(B2)
```

In cell F2 type: =TRIM(B2)

This cleans ' ravi kumar ' to 'ravi kumar' (spaces removed).

Copy down for all rows. TRIM is used constantly in real data work!

■ PROPER — Converts text to Proper Case (First Letter Capital)

```
=PROPER(TRIM(B2))
```

In cell F2, combine both: =PROPER(TRIM(B2))

First TRIM removes spaces, then PROPER fixes capitalization.

Result: 'Ravi Kumar' — clean and professional!

■ MID — Extracts text from the MIDDLE of a string (starting at any position)

```
=MID(text, start_position, how_many_characters)
```

Example: =MID("EMP001", 4, 3) gives '001' (start at position 4, take 3 chars)

For employee number from ID: =MID(A2, 4, 3)

MID is powerful — can extract city codes, account numbers, etc.

■ RIGHT — Extracts characters from the RIGHT end of text

```
=RIGHT(text, how_many_characters)
```

Example: =RIGHT("SALES-01", 2) gives '01' (last 2 characters)

For department number: =RIGHT(C2, 2)

Similar formula: LEFT(text, chars) extracts from the left side.

■ CONCATENATE — Joins (combines) multiple cells or text into one

```
=CONCATENATE(A2, " - ", F2, " - ", C2)
```

Or use the & shortcut: =A2 & " - " & F2 & " - " & C2

Result: 'EMP001 - Ravi Kumar - SALES-01'

Use for: full address, invoice descriptions, combined labels.

■ *REAL WORLD: When data is imported from another software (like Tally export to Excel, or bank statement downloads), the names are often messy — extra spaces, wrong case. Before doing any analysis, accountants and data analysts always run TRIM and PROPER first. This is called 'data cleaning' and it takes 30-40% of a real analyst's time!*

■ TASK 4 — Date & Time Calculations (≈30 minutes)

■ Scenario

You manage a team of employees. You need to calculate: how old each employee is, how many years they have worked, when their loan EMI ends, and what day of the week their birthday falls on this year. Date formulas are used in EVERY business — for ageing reports, loan tracking, GST deadlines, and HR records!

■ Step 1: Type This Data (Sheet name: Date Calculations)

A	B	C	D	E
Employee	**Date of Birth**	**Date of Joining**	**Loan Start Date**	**Loan Months**
Ravi Kumar	15-03-1995	01-06-2020	01-01-2024	24
Priya Singh	22-07-1998	15-03-2021	01-04-2024	12
Aman Verma	08-11-1992	01-01-2019	01-07-2023	36
Sunita Rao	30-09-2000	01-09-2022	01-10-2024	18
Rohit Sharma	14-05-1990	15-06-2018	01-03-2024	24

■ Step 2: Create These Answer Columns (Column G onwards)

G = Age (Years) | H = Years of Service | I = Loan End Date | J = Birthday Day Name | K = Today's Date

■ Step 3: Learn These 5 Date Formulas

■ TODAY — Returns today's date automatically — updates every day

```
=TODAY()
```

In cell K2 type: =TODAY()

This always shows the current date — no need to type it manually.

Every time you open the file, it shows today's fresh date automatically.

Used in: ageing reports, overdue tracking, deadline calculations.

■ DATEDIF — Calculates difference between two dates in Years, Months, or Days

```
=DATEDIF(start_date, end_date, "Y")
```

For Age in Years — in cell G2: =DATEDIF(B2, TODAY(), "Y")

"Y" = complete years | "M" = complete months | "D" = days

For Years of Service in H2: =DATEDIF(C2, TODAY(), "Y")

Note: DATEDIF is a hidden formula — type it manually, it will work!

■ EOMONTH — Finds the LAST DATE of a month (N months in the future)

```
=EOMONTH(start_date, number_of_months)
```

For Loan End Date in I2: =EOMONTH(D2, E2)

D2 = loan start date | E2 = number of months (24)

Result: the last day of the month when loan ends.

Used for: EMI end dates, subscription expiry, GST due dates.

■ WEEKDAY — Tells you what DAY of the week a date falls on (1=Sunday, 2=Monday...)

```
=WEEKDAY(date, 2)
```

=WEEKDAY(B2, 2) gives a number: 1=Monday, 2=Tuesday... 7=Sunday

To show the DAY NAME, combine with CHOOSE:

=CHOOSE(WEEKDAY(B2,2),"Mon","Tue","Wed","Thu","Fri","Sat","Sun")

Use this in J2 to show which day each employee was born on!

■ NOW — Returns the current DATE and TIME together

```
=NOW()
```

Type =NOW() in any cell — it shows: 12-07-2025 14:35:22

Updates every time the sheet recalculates.

Useful for: timestamping when a report was last updated.

Difference from TODAY(): NOW() includes time, TODAY() is date only.

■ TALLY CONNECTION: In Tally, the Ageing Analysis report uses DATEDIF logic to classify invoices into 0-30, 31-60, 61-90 day buckets. The BRS (Bank Reconciliation) uses date comparison. GST due dates use EOMONTH. All the date logic you learned today powers Tally's reports behind the scenes!

■ FORMULA QUICK REFERENCE CARD

Formula	What it does	Simple Example	Task
AVERAGE	Mean of numbers	`=AVERAGE(B2:E2)`	Task 1
MAX	Highest value	`=MAX(G2:G9)`	Task 1
MIN	Lowest value	`=MIN(G2:G9)`	Task 1
RANK	Position in a list	`=RANK(G2,$G$2:$G$9,0)`	Task 1
SUMIF	Sum with condition	`=SUMIF(B2:B13,F2,D2:D13)`	Task 2
COUNTIF	Count with condition	`=COUNTIF(B2:B13,F2)`	Task 2
AVERAGEIF	Average with condition	`=AVERAGEIF(B2:B13,F2,D2:D13)`	Task 2
TRIM	Remove extra spaces	`=TRIM(B2)`	Task 3
PROPER	Proper Case text	`=PROPER(B2)`	Task 3
MID	Extract middle chars	`=MID(A2,4,3)`	Task 3
RIGHT	Extract right chars	`=RIGHT(C2,2)`	Task 3
CONCATENATE	Join text together	`=CONCATENATE(A2," ",B2)`	Task 3
TODAY	Current date	`=TODAY()`	Task 4
DATEDIF	Days/months/years between dates	`=DATEDIF(B2,TODAY(),"Y")`	Task 4
EOMONTH	Last day of future month	`=EOMONTH(D2,E2)`	Task 4
WEEKDAY	Day number of week	`=WEEKDAY(B2,2)`	Task 4
NOW	Current date and time	`=NOW()`	Task 4